

Introducing Bootstrap 4



Jörg Krause

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Jörg Krause
Berlin, Germany

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*This book is for everyone making their first steps into the world of software.
Or for those wish to bring their knowledge up to date.*

*The future of software development may be the web, the cloud, or where ever.
In any case, it is not on an isolated local system.*

*This book is part of a project, a set of textbooks, to help you
fight the challenges of web development.*

The subjects are not necessarily new, but do form a unique subject.

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About the Author



Jörg Krause works as a trainer, consultant and software developer for major companies worldwide. Build on the experience of 25 years of work with web and many, many large and small projects. Jörg believes it is especially important to have solid foundations. Instead of always running to create the latest framework, many developers would be better advised to create and provide a robust foundation.

Jörg has written over 50 titles in the renowned and prestigious specialist publishers in German and English, including some bestsellers. Anyone who wants to learn this subject compact and fast, its right here. On his website www.joergkrause.de much more information can be found.

About the Technical Reviewer



Massimo Nardone has more than twenty-two years experience in security, web/mobile development, cloud, and IT architecture. His true IT passions are in the security and android field.

He has been programming and teaching how to program with android, Perl, PHP, Java, VB, Python, C/C++, and MySQL for more than twenty years. He holds a Master of Science degree in Computing Science from the University of Salerno, Italy.

He has worked as a project manager, software engineer, research engineer, chief security architect, information security manager, PCI/SCADA auditor, and senior lead IT security/cloud/SCADA architect for many years.

His technical skills include: security, android, cloud, Java, MySQL, Drupal, Cobol, Perl, web, and mobile development; MongoDB, d3, Joomla, Couchbase, C/C++, WebGL, Python, Pro Rails, django CMS, Jekyll, Scratch, and many others.

He is currently working as the chief information security officer (CISO) for Cargotec Oyj.

He worked as visiting lecturer and supervisor for exercises at the Networking Laboratory of the Helsinki University of Technology (Aalto University). He holds four international patents (PKI, SIP, SAML, and Proxy areas).

Massimo has reviewed more than forty IT books for different publishing company, and he is the coauthor of *Pro Android Games* (Apress, 2015).

Introduction

Bootstrap 4 – the new CSS Framework

This book describes in a compact and clear form the CSS Framework Bootstrap 4. Bootstrap was originally designed, around 2010, on Twitter and was from the outset developed with the idea of “Mobile First” – first for mobile devices . It has since grown to become one of the best and most popular frameworks for the design of websites. Accompanied by a large number to bootstrap engaging design templates – so-called themes – it offers secure and robust style definitions for everyday tasks.

Target Audience

This book is aimed at beginners and web developers who are new to the web world. Bootstrap serves mainly the front-end developer. It is a series of CSS instructions and these additional JavaScript (ECMAScript) libraries, apply where CSS alone is not enough.

Maybe you are also a web designer, who discovered Bootstrap as an excellent way to upgrade your web pages with dynamic elements. Here, you are dealing with texts, with forms, with the presentation of database content, i.e. everything that constitutes a dynamic web site. Then you will need to know one of this band, namely for the creation of a professional user interface, in particular a clear form.

In any case, I tried not to ask an prerequisites or conditions to the reader. You do not need to be a computer scientist, not in perfect command of language, don't need to know rocket science. No matter in what context they have encountered on Bootstrap, you will be able to read this text.



Bootstrap For all examples to understand, you need a working environment for creating web pages. This can be Ruby on Rails, PHP, ASP.NET or Node.js. I recommend node.js for an easy start. It's a nice way to get Bootstrap up and running.

If you have accidentally found this text and cannot do anything with the term “Bootstrap”, read it anyway. You will be learning one of the most modern techniques of web development and the future belongs to the circle of excellent developers who can be build good- looking and device-independent sites.

What You Should Know Readers of this series have hardly any requirements. Some HTML cannot harm and who already have seen a static HTML page (the source code, of course) is certainly good and pure. I assume that you have at least a current operating system, on which you will find an editor with which you can create web pages.

Examples

You can find the sample project to this book on Github:

- <https://github.com/joergkrause/Bootstrap4-Book>

The project uses sample files contain the pure HTML. It works best with Visual Studio, but can largely be used independently.

As You Can Read This Text

I will not dictate how you should read this text. In the first draft of the structure, I have tried several variations and found that there exists no ideal form. If I were to orientate myself to the various types of applications, the text would divided into several chapters that are not interrelated. The one or the other readers would then be annoyed that he put a lot of money for a book, where only a fifth is usable. This band solves the problem by reducing it to a very small issue is focused and no “blah-blah” for the inflation of the volume is.

Beginners should read the text as a narrative from the first to the last page. Those who are already somewhat familiar, the less interesting sections for him, he can safely skip sections. If references are required, I have included the corresponding appropriate cross- references.

Realizations

The theme is set not technically easy to master, because scripts are often too extensive and it would be nice if you could support the best optical reading form. I have therefore included extra line breaks used to aid readability, but have no place in the editor of their development environment.

In general, each program code is set to a non-proportional font. In addition, scripts have line numbers: :

```
1  body {  
2    color: black;  
3  }
```

If you find you need to enter something in the prompt or in a dialog box, this part of the statement is in bold:

```
$ bower install bootstrap
```


The first character is the prompt and is not entered. I use in the book, the Linux prompt, the bash shell. The commands will work, without any exception, unchanged even on Windows, only then STOP C:> or something similar at the beginning of the line.

Expressions and command lines are sometimes peppered with all types of characters, and in almost all cases, it depends on each character. Often, I'll discuss the use of certain characters in precisely such an expression. Then the “important” characters with line breaks alone and also in this case, line numbers line numbers are used to reference the affected symbol in the text exactly (note the : character in line 2):

```
1  a.test {
2      :hover {
3          color: red
4      }
5  }
```

The font is non-proportional, so that the characters are countable and opening and closing parentheses are always among themselves.

Symbols

To facilitate the orientation in the search for a solution, there is a whole range of symbols that are used in the text.



Tip This is a tip



Information This is an information



Warning This is a warning
